

Python API Summary

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SYNOPSYS[®]

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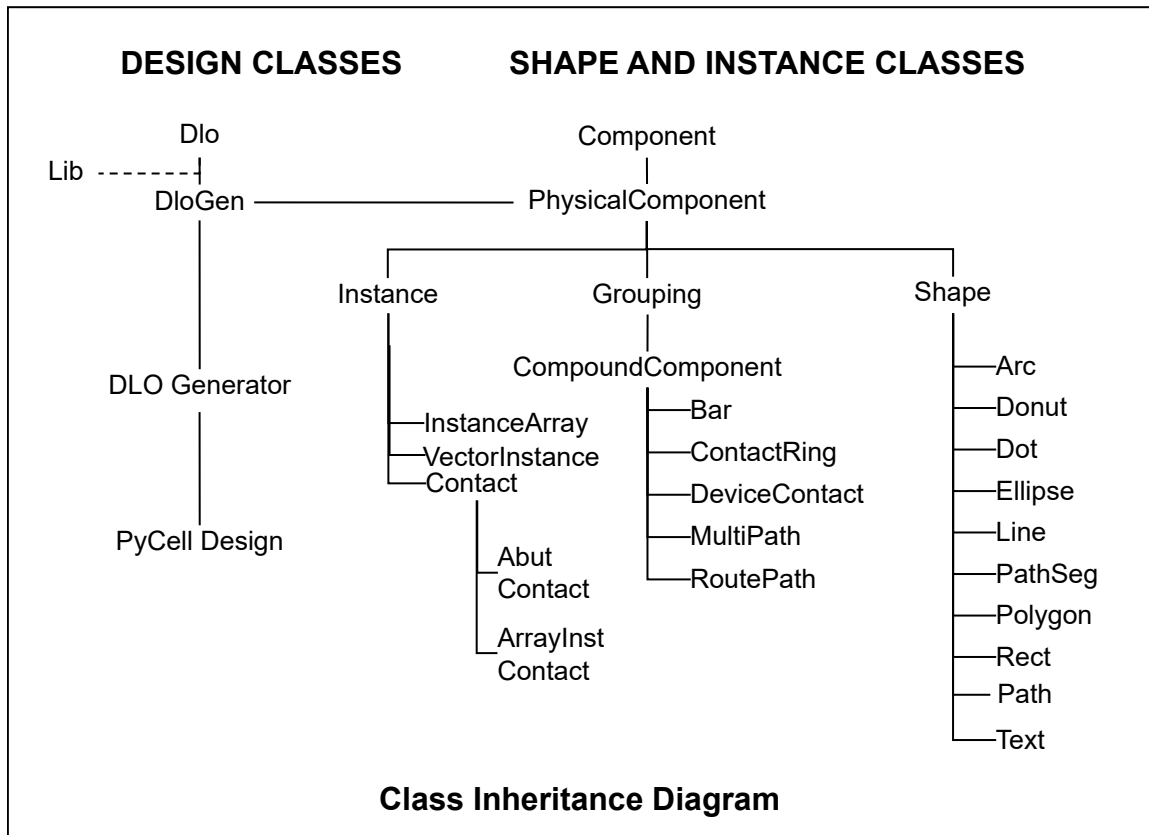
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Class Organization



Technology Classes

Grid, Layer, LayerMaterial, PhysicalRule, RuleProperty, ShapeFilter, Tech

Connectivity Classes

BundleNet, BundleTerm, BusNet, BusTerm, InstPin, InstTerm, Net, Pin, RouteTarget, ShapeRef, SignalType, Term, TermType, Topology

Basic Geometric Classes

Box, Direction, Font, GapStyle, Location, Orientation, PathStyle, Point, Range, Segment, Transform

Utility Classes

AttrDict, CDF, cFloat, CrossOver, DPL,, DrcSummary, Fill, Grid, Log, Marker, NameMapper, Numeric, OrderedDict, ParamArray, ParamDictSpec, ParamSpecArray, PointList, PropSet, SnapType, Unique, ViaParam

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Class Methods

Design Classes

Dlo

exists(), findComp(), findCompRef(), findPin(), findTerm(), getBBox(), getCellType(), getComps(), getLeafComps(), getLib(), getLpps(), getName(), getParams(), getPins(), getProps(), getTermOrder(), getTerms(), getViewType(), isSubMaster(), isSuperMaster(), save(), saveAs(), setCustomLocation(), setTermOrder()

DloGen

addPin(), addTerm(), currentDloGen(), dbu2uu(), dbu2uuArea(), fgAnd(), fgDeriveLayer(), fgNot(), fgOr(), fgSize(), fgXor(), getComp(), getInsts(), getNets(), getShapes(), makeCompName(), makeGrouping(), makeNetName(), makePinName(), makePinNameSpecNum(), makeTermName(), mirrorX(), mirrorY(), moveBy(), moveTo(), moveTowards(), rotate90(), rotate180(), rotate270(), setOrigin(), transform(), uu2dbu(), uu2dbuArea(), withNewDlo()

PyCell author provided methods:

defineParamSpecs(), genLayout(), genTopology(), setupParams(), sizeDevices()

Lib

attachTechLib(), close(), create(), definePcell(), getName(), getPath(), getProps(), getTech(), getTechFilePath(), getTechLibName(), installTechFile(), loadPcells(), open(), updatePcell()

Shape and Instance Classes

PhysicalComponent

abut(), alignEdge(), alignEdgeToPoint(), alignLocation(), alignLocationToPoint(), clone(), destroy(), find(), fgAbut(), fgAddEnclosingPolygon(), fgAnd(), fgDeriveLayer(), fgEnclose(), fgExtend(), fgFill(), fgMerge(), fgMinSpacing(), fgNot(), fgOr(), fgPlace(), fgSize(), fgXor(), getBBox(), getCompOwner(), getDlo(), getLocationPoint(), getName(), getProps(),

`getSpacing()`, `keepRelativePosition()`, `makeArray()`, `mirrorX()`, `mirrorY()`, `moveBy()`,
`moveTo()`, `moveTowards()`, `overlaps()`, `place()`, `rotate90()`, `rotate180()`, `rotate270()`,
`setCustomLocation()`, `setName()`, `snap()`, `snapX()`, `snapY()`, `snapTowards()`, `transform()`

Instance

`clone()`, `find()`, `findInstPin()`, `findInstTerm()`, `flatten()`, `getBit()`, `getCompRefs()`,
`getDefaultParams()`, `getDloName()`, `getInstPins()`, `getInstTerms()`, `getMaster()`,
`getName()`, `getNumBits()`, `getOrientation()`, `getOrigin()`, `getParams()`, `getParamSpecs()`,
`getTransform()`, `instPin()`, `setConnectivity()`, `setMaster()`, `setName()`, `setOrientation()`,
`setOrigin()`, `setParams()`, `setTransform()`

VectorInstance

`clone()`, `find()`, `flatten()`, `getBaseName()`, `getBit()`, `getCompRefs()`, `getNumBits()`,
`getStartIndex()`, `getStopIndex()`, `setBaseName()`, `setIndexRange()`

InstanceArray

`clone()`, `find()`, `flatten()`, `getCompRefs()`, `getDX()`, `getDY()`, `getMemberBBox()`,
`getMemberRefs()`, `getMemberTransform()`, `getNumCols()`, `getNumRows()`, `setDX()`,
`setDY()`, `setName()`, `setNumCols()`, `setNumRows()`

Contact

`clone()`, `getLayer1()`, `getLayer2()`, `getLayers()`, `getNumCuts()`, `getNumHVCuts()`,
`getRefBox()`, `setMinCuts()`, `stretch()`, `stretchTo()`, `stretchToCoord()`

AbutContact

`clone()`, `setAbutViaSpaceFactor()`

ArrayInstContact

`clone()`

ContactRing

`chop()`, `clone()`, `getContact()`

Grouping

`find()`, `add()`, `clone()`, `destroy()`, `flatten()`, `getBBox()`, `getComp()`, `getComps()`,
`getLeafComps()`, `getName()`, `isPersistent()`, `makePersistent()`, `mirrorX()`, `mirrorY()`,
`moveBy()`, `moveTo()`, `moveTowards()`, `remove()`, `rotate90()`, `rotate180()`, `rotate270()`,
`setName()`, `transform()`, `ungroup()`

CompoundComponent

clone(), destroy(), flatten(), isLocked(), lock(), mirrorX(), mirrorY(), moveBy(), moveTo(), moveTowards(), rotate90(), rotate180(), rotate270(), snap(), snapX(), snapY(), snapTowards(), transform(), ungroup(), unlock()

Bar

addContact(), clearContacts(), clone(), extendTo(), getContacts(), getDirection(), getLayer(), getRect(), getRoutePathIntersectBox(), stretchTo(), stretchToCoord(), trim()

DeviceContact

clone(), getLayer1(), getLayer2(), getNumHVCuts(), getRect1(), getRect2(), getViaBBox(), getViaLayer(), stretch(), stretchToCoord(), stretchToPoint()

MultiPath

clone(), createEnclosureSubpath(), createOffsetSubpath(), createSubrectangles(), genJustifyPathPoints(), getChoppedSubpathPointLists(), getMasterPathName(), getSubpathPoints(), mirrorX(), mirrorY(), moveBy(), moveTo(), moveTowards(), rotate90(), rotate180(), rotate270(), snap(), setChopBoxes(), setChoppable(), snapX(), snapY(), snapTowards(), transform()

RoutePath

Connect(), CShape(), FlightLine(), LShape(), StraightLine(), StraightLineToBar(), ZShape(), checkLayerWidth(), clone(), findAdjacentInterconnectLayer(), getLayer()

PRBoundary

find(), getBBox(), getEdgeNames(), getNumEdges(), getPoints(), setEdges()

Shape Classes

getBBox(), getColorMask(), getLayer(), getName(), getNet(), getPin(), setColorMask(), setLayer(), setName(), setNet()

Arc

Arc(), clone(), getEllipseBBox(), getStartAngle(), getStopAngle(), setEllipseBBox(), setStartAngle(), setStopAngle()

Donut

Donut(), clone(), getCenter(), getEndAngle(), getHoleBBox(), getHoleRadius(), getRadius(), getStartAngle(), setAngles(), setCenter(), setEndAngle(), setHoleRadius(), setRadius(), setStartAngle()

Dot

Dot(), clone(), getHeight(), getOrigin(), getWidth(), setHeight(), setOrigin(), setWidth()

Ellipse

Ellipse(), clone(), genPolygonPoints(), getEndAngle(), getStartAngle(), setAngles(), setBBox(), setEndAngle(), setStartAngle()

Line

Line(), clone(), getNumPoints(), getPoints(), setPoints()

Path

Path(), clone(), getBeginExt(), getBoundary(), getEdgeLength(), getEndExt(), getNumPoints(), getPoints(), getStyle(), getWidth(), isOrthogonal(), setBeginExt(), setEndExt(), setPoints(), setStyle(), setWidth()

PathSeg

PathSeg(), clone(), getBoundary(), getPoints(), getWidth(), isOrthogonal(), setPoints(), setWidth()

Polygon

Polygon(), clone(), getEdgeLength(), getNumPoints(), getPoints(), isOrthogonal(), setPoints(), subdivide()

Rect

Rect(), clone(), expand(), expandDir(), expandToGrid(), fillBBoxWithRects(), fillDiagBoxWithDiagRects(), fillDiagBoxWithRects(), getBottom(), getCoord(), getHeight(), getLeft(), getRight(), getTop(), getWidth(), setBBox(), setBottom(), setCoord(), setLeft(), setRight(), setTop()

Text

Text(), clone(), getAlignment(), getHeight(), getOrientation(), getOrigin(), getText(), hasOverbar(), isDrafting(), isVisible(), setAlignment(), setDrafting(), setHeight(), setOrientation(), setOrigin(), setOverbar(), setText(), setVisible()

AttrDisplay

getAttribute(), getObject()

Via

getBBox(), getColorMask(), getLayer1(), getLayer2(), getMaster(), getName(), getNet(), getOrientation(), getOrigin(), getTransform(), getViaDefName(), setColorMask(), setName(), setNet(), setOrientation(), setOrigin(), setTransform()

StdVia

getCutLayer(), getImplantLayer1(), getImplantLayer2(), getParams(), setParams()

CustomVia

getParams(), setParams()

ViaDef

getName(), getLayer1(), getLayer2()

StdViaDef

getImplantLayer1(), getImplantLayer2(), getParams()

CustomViaDef

getLibName(), getCellName(), getViewName(), getParams()

PhysicalCompRef

abut(), alignEdge(), alignEdgeToCoord(), alignEdgeToPoint(), alignLocation(), alignLocationToPoint(), getBBox(), getLocationPoint(), getName(), getParentPathName(), getParentPathTransform(), getPathName(), getProps(), getSpacing(), getTopInst(), keepRelativePosition(), mirrorX(), mirrorY(), moveBy(), moveTo(), moveTowards(), overlaps(), place(), rotate90(), rotate180(), rotate270(), snap(), snapTowards(), snapX(), snapY(), transform()

GroupingRef

getCompRefs()

InstanceRef

getCompRefs(), getMaster(), getParams()

InstanceArrayRef

getMemberRefs()

VectorInstanceRef

getCompRefs()

ShapeRef

getBBox(), getInst(), getInstPin(), getLayer(), getName(), getNetRef(), getPinRef(), getTransform()

TextRef

getText()

Basic Geometric Classes

Point

areColinearPoints(), copy(), getCoord(), getSpacing(), getX(), getY(), isBetween(), invalid(), isValid(), place(), set(), setCoord(), setX(), setY(), snap(), snapX(), snapY(), snapTowards(), toDiagAxes(), toOrthogAxes(), transform()

Range

alignEdge(), alignEdgeToCoord(), compareTrueCenter(), contains(), containsCoord(), expand(), expandDir(), findAlignEdgeOffset(), fitSubranges(), fix(), getCenter(), getCoord(), getLeft(), getRight(), getWidth(), hasNoWidth(), init(), intersect(), isInverted(), isNormal(), limit(), merge(), mergeCoord(), moveBy(), overlaps(), removeRegion(), set(), setCenter(), setCoord(), setDimension(), setLeft(), setRight(), setWidth()

Segment

addOffsets(), contains(), containsPoint(), extrapIntersect(), genJustifySegment(), getDeltaX(), getDeltaY(), getDir(), getHead(), getPosition(), getTail(), hasIntersection(), intersect(), isCoincident(), isHorizontal(), isOrthogonal(), isParallel(), isVertical(), moveBy(), reverse(), set(), setHead(), setTail(), transform()

Box

abut(), alignEdge(), alignEdgeToCoord(), alignEdgeToPoint(), alignLocation(), alignLocationToPoint(), centerLeft(), centerRight(), centerCenter(), contains(), containsPoint(), expand(), expandDir(), expandForMinArea(), expandForMinWidth(), expandToGrid(), fix(), getArea(), getBottom(), getCenter(), getCenterX(), getCenterY(), getCoord(), getDimension(), getHeight(), getLeft(), getLocationPoint(), getPoints(), getRange(), getRangeX(), getRangeY(), getRight(), getSpacing(), getTop(), getWidth(), hasNoArea(), init(), intersect(), isInverted(), isNormal(), limit(), lowerCenter(), lowerLeft(), lowerRight(), merge(), mergePoint(), mirrorX(), mirrorY(), moveBy(), moveTo(), moveTowards(), overlaps(), place(), removeRegion(), rotate90(), rotate180(), rotate270(), set(), setBottom(), setCenter(), setCenterX(), setCenterY(), setCoord(), setDimension(), setHeight(), setLeft(), setLocationPoint(), setRange(), setRangeX(), setRangeY(), setRight(), setTop(), setWidth(), snap(), snapX(), snapY(), snapTowards(), transform(), upperCenter(), upperLeft(), upperRight()

Direction

containsComponent(), extend(), getJustifiedDir(), getMembers(), getPrimaryDirs(), is1Dimension(), isHalfLine(), isFullLine(), isPrimary(), isXDir(), isYDir(), mapXDirToYDir(), mapYDirToXDir(), mirrorX(), mirrorY(), opposite(), perpendicular(), rotate90(), rotate180(), rotate270(), transform()

Location

mirrorX(), mirrorY(), rotate90(), rotate180(), rotate270(), transform()

Orientation

concat(), getRelativeOrient()

Transform

concat(), invert(), mirrorX(), mirrorY(), rotate90(), rotate180(), rotate270()

Technology Classes

Layer

getAttrs(), getGridResolution(), getLayerAbove(), getLayerBelow(), getLayerName(),
getLayerNumber(), getMaterial(), getPurposeName(), getPurposeNumber(),
getRoutingDir(), isAbove(), isMaskLayer()

LayerMaterial

include()

ShapeFilter

exclude(), excludeTexts(), include(), isIncluded()

DeviceContext

getLayers(), getName(), getRuleSubstitutions(), isEmpty()

Ruleset

getAncestor(), getName()

Tech

conditionalRuleExists(), dbu2uu(), dbu2uuArea(), deviceContextExists(),
electricalRuleExists(), get(), getActiveDeviceContext(), getActiveRuleset(),
getCustomViaDef(), getCustomViaDefNames(), getDeviceContexts(), getElectricalRule(),
getGridResolution(), getIntermediateLayers(), getLayer(), getMaskColorNames(),
getMosfetParams(), getOxideParams(), getPhysicalRule(), getRulesets(),
getSantanaLayerNames(), getSantanaPurposeNames(), getStdViaDef(),
getStdViaDefNames(), getTechParams(), getUserUnits(), getValidLayers(), id(), name(),
physicalRuleExists(), revision(), rulesetExists(), uu2dbu(), uu2dbuArea(), version()

Point List Geometry Operations

Point List Geometry Methods

poAnd(), poInside(), poNot(), poOr(), poOutside(), poSize(), poStraddle(), poXor()

Collection

addPointLists(), addPointListsByObjects(), createPointListsToShapes(), getPointLists()

Connectivity Classes

Net

addShape(), destroy(), find(), findCreate(), getBit(), getName(), getNumBits(), getShapes(), getSignalType(), getInstPins(), getInstTerms(), getPins(), getTerm(), getVias(), isGlobal(), removeShape(), setGlobal(), setName(), setNetOverride(), setSignalType()

BusNet

find(), getBaseName(), getBit(), getIndexStep(), getNumBits(), getStartIndex(), getStopIndex(), setBaseName(), setIndexRange()

BundleNet

find(), getMember(), getMembers(), getName(), getNumMembers(), getRepeat()

Pin

addInstPin(), addShape(), destroy(), find(), getAccessDir(), getAuthorPrefLayer(), getBBox(), getLayers(), getName(), getNet(), getPrefLayer(), getPrefLayerPropName(), getShapes(), getTerm(), getTopLayer(), removeShape(), setAccessDir(), setAuthorPrefLayer(), setName(), setTerm()

Term

destroy(), find(), findCreate(), getBit(), getMustJoinTerms(), getName(), getNet(), getNumBits(), getPins(), getTermType(), setMustJoin(), setName(), setNetOverride(), setTermType()

BusTerm

find(), getBaseName(), getBit(), getIndexStep(), getNumBits(), getStartIndex(), getStopIndex(), setBaseName(), setIndexRange()

BundleTerm

find(), getMember(), getMembers(), getName(), getNumMembers()

InstPin

find(), getAuthorPrefLayer(), getBBox(), getInst(), getInstTerm(), getLayers(), getNet(), getPinAccessDir(), getPinName(), getPrefLayer(), getShapeRefs(), getTermName(), getTermType(), getTopLayer()

InstTerm

find(), findInstPin(), getBit(), getInst(), getInstPins(), getMustJoinInstTerms(), getNet(), getNumBits(), getTermName(), getTermType(), setNet()

Topology

getName(), setName()

RouteTarget

getBBox(), getChosenAccessDir(), getChosenAccessPoint(), getChosenBox(), getChosenLayer(), getLayers(), getName(), getPrefLayer(), getRoutePathIntersectBox(), getRoutePathLayer(), hasLayer(), isSingleBox(), isValid(), needsContact()

Connectivity Reference Classes

BlockObjectRef

getProps()

NetRef

getBit(), getName(), getNumBits(), getPinRefs(), getShapeRefs(), getSignalType(), getTermRef(), getViaRefs(), isGlobal()

TermRef

getName(), getNetRef(), getPinRefs(), getTermType()

PinRef

getAccessDir(), getName(), getNetRef(), getShapeRefs(), getTermRef()

Utility Classes

ParamArray

add(), get(), has_key(), iteritems(), iterkeys(), itervalues(), remove(), reset(), set(), setFromSpecs(), update()

ParamSpecArray

`add()`, `has_key()`, `iteritems()`, `iterkeys()`, `intervalues()`, `verify()`

ViaParam

`hasDefault()`, `items()`, `iteritems()`, `keys()`, `iterkeys()`, `setDefault()`

PropSet

`clear()`, `empty()`, `get()`, `getProp()`, `has_key()`, `items()`, `iteritems()`, `iterkeys()`, `intervalues()`, `keys()`, `update()`, `values()`

PointList

`addOffsets()`, `compress()`, `containsPoint()`, `copy()`, `deepcopy()`, `genJustifyPoints()`, `getArea()`, `getBBox()`, `hasExtraPoints()`, `isOrthogonal()`, `isSelfIntersecting()`, `mirrorX()`, `mirrorY()`, `moveBy()`, `moveTo()`, `moveTowards()`, `onEdge()`, `rotate90()`, `rotate180()`, `rotate270()`, `transform()`

Log

`debug()`, `error()`, `info()`, `test()`, `warning()`

Unique

`Name()`

NameMapper

`map()`

Grid

`getSize()`, `getSnapType()`, `getXSize()`, `getYSize()`, `setSize()`, `setSnapType()`, `setXSize()`, `setYSize()`, `snap()`, `snapX()`, `snapY()`

SnapType

`snap()`, `ceil()`, `floor()`, `round()`, `round_ceil()`, `trunc()`

Fill

`fillPhysCompWithInstArrays()`,
`fillPhysCompWithLargeArraysOfInstArrays()`,`fillPhysCompWithLargeArraysOfRects()`,
`fillPhysCompWithRects()`

Numeric

`scaleFormat()`

CDF

`forCell()`

AttrDict

`clear()`, `get()`, `has_key()`, `items()`, `iteritems()`, `iterkeys()`, `itervalues()`, `keys()`, `pop()`, `popitem()`, `setdefault()`, `update()`, `values()`

OrderedDict

`clear()`, `copy()`, `index()`, `insert()`, `items()`, `iteritems()`, `iterkeys()`, `itervalues()`, `keys()`, `pop()`, `popitem()`, `rename()`, `reverse()`, `setdefault()`, `setitems()`, `setkeys()`, `setvalues()`, `sort()`, `update()`, `values()`

ParamDictSpec

`match()`, `optional()`, `required()`

CrossOver

`genPath1Polygon()`, `genPath2Polygon()`, `getDistance()`, `getPath1Extends()`, `getPath1Points()`, `getPath2Extends()`, `getPath2Points()`, `getSpace()`, `getWidth1()`, `getWidth2()`, `getWidth45()`, `set()`

Marker

`addObject()`, `clone()`, `getBBox()`, `getMsg()`, `getObjects()`, `getPoints()`, `getSeverity()`, `getShortMsg()`, `getTool()`, `isClosed()`, `isVisible()`, `setIsClosed()`, `setIsVisible()`, `setMsg()`, `setPoints()`, `setSeverity()`, `setShortMsg()`, `setTool()`, `removeObject()`

DrcSummary

`checkedRules()`, `errorNumber()`, `getRuleList()`, `totalErrorNumber()`

AppDef

`destroy()`, `get()`, `set()`

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Global Functions

`fgDrc()`, `stretchHandle()`, `stretchHandleCustom()`, `autoAbutment()`